

Ing. Tomáš Martinovič, Ph.D.

Senior Researcher · IT4Innovations, VSB – Technical University of Ostrava

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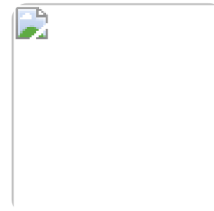
🐙 <https://github.com/Nithador>

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Employment History

- 2022-present ■ **Senior Researcher.** IT4Innovations, VSB – Technical University of Ostrava.
- 2019-2022 ■ **Researcher.** IT4Innovations, VSB – Technical University of Ostrava.
- 2016-2018 ■ **Research Assistant.** IT4Innovations, VSB – Technical University of Ostrava.
- 2014-2015 ■ **Junior Data Analyst.** IT4Innovations, VSB – Technical University of Ostrava.

Education

- 2015-2018 ■ **Ph.D., Computational Sciences**, VSB – Technical University of Ostrava.
Thesis title: *Tools for time series analysis of nonlinear dynamical systems.*
- 2012-2015 ■ **Ing., Quantitative Methods in Economy**, Masaryk University.
Thesis title: *Investigating uncertainty in the estimates of potential output and the NAIRU.*
- 2009-2012 ■ **Bc., Mathematical Methods in Economics**, Prague University of Economics and Business.
Thesis title: *Operation research applications in public sector.*

Research Publications

Journal Articles

- 1 Desalegn Chala et al. "Prototype biodiversity digital twin: crop wild relatives genetic resources for food security." *Research Ideas and Outcomes*, 2024. DOI: [10.3897/rio.10.e125192](https://doi.org/10.3897/rio.10.e125192).
- 2 Jürgen Groeneveld et al. "Prototype Biodiversity Digital Twin: honey bees in agricultural landscapes." *Research Ideas and Outcomes*, 2024. DOI: [10.3897/rio.10.e125167](https://doi.org/10.3897/rio.10.e125167).
- 3 Muhammad Iqbal et al. "Demonstration of Sensitive Analysis and Optical Soliton Patterns in a (4+1) Dimensional Boiti-Leon-Manna Pempinelli Equation: Dynamic Insights into Bifurcation, Chaotic Behavior." *Chaos Theory and Applications*, 2025. DOI: [10.51537/chaos.1518307](https://doi.org/10.51537/chaos.1518307).
- 4 Adil Jhangeer et al. "A new technique for characterizing nonlinear dynamical system behavior using self-organizing maps and clustering." *Ain Shams Engineering Journal*, 2026. DOI: [10.1016/j.asej.2026.104206](https://doi.org/10.1016/j.asej.2026.104206).

- 5 Marek Lampart and Tomas Martinovic. "A survey of tools detecting the dynamical properties of one-dimensional families." *Advances in Electrical and Electronic Engineering*, 2017. DOI: [10.15598/aeed.v15i2.2314](https://doi.org/10.15598/aeed.v15i2.2314).
- 6 Stéphane Louise et al. "Modeling and implementing an earthquake and tsunami event-triggered, time-constrained impact assessment workflow." *The International Journal of High Performance Computing Applications*, 2025. DOI: [10.1177/10943420251343794](https://doi.org/10.1177/10943420251343794).
- 7 Tomas Martinovic and Jan Fulneczek. "Fast Algorithm for Contactless Partial Discharge Detection on Remote Gateway Device." *IEEE Transactions on Power Delivery*, 2022. DOI: [10.1109/TPWRD.2021.3104746](https://doi.org/10.1109/TPWRD.2021.3104746).
- 8 T. Martinovič. "Alternative approaches of evaluating the 0-1 test for chaos." *International Journal of Computer Mathematics*, 2020. DOI: [10.1080/00207160.2019.1701662](https://doi.org/10.1080/00207160.2019.1701662).
- 9 Tomáš Martinovič. "Chaotic behaviour of noisy traffic data." *Mathematical Methods in the Applied Sciences*, 2018. DOI: [10.1002/mma.4234](https://doi.org/10.1002/mma.4234).
- 10 Tomáš Martinovič and Georg Zitzlsberger. "Highly scalable algorithm for computation of recurrence quantitative analysis." *The Journal of Supercomputing*, 2019. DOI: [10.1007/s11227-018-2350-5](https://doi.org/10.1007/s11227-018-2350-5).
- 11 Andrii Patrikei et al. "Essential time series characteristics for human motion analysis based on Self-Organizing Map clustering." *Acta Gymnica*, 2026. DOI: [10.5507/ag.2026.004](https://doi.org/10.5507/ag.2026.004).
- 12 Joan Planas-Iglesias et al. "AggreProt: a web server for predicting and engineering aggregation prone regions in proteins." *Nucleic Acids Research*, 2024. DOI: [10.1093/nar/gkae420](https://doi.org/10.1093/nar/gkae420).
- 13 Simon Rolph et al. "Prototype Digital Twin: Recreation and biodiversity cultural ecosystem services." *Research Ideas and Outcomes*, 2024. DOI: [10.3897/rio.10.e125450](https://doi.org/10.3897/rio.10.e125450).
- 14 Allan Souza et al. "Fragmented Networks: Challenges in communication and cohesion of European Biodiversity Research Infrastructures." *Biodiversity Data Journal*, 2025. DOI: [10.3897/BDJ.13.e148079](https://doi.org/10.3897/BDJ.13.e148079).
- 15 Franziska Taubert et al. "Prototype Biodiversity Digital Twin: grassland biodiversity dynamics." *Research Ideas and Outcomes*, 2024. DOI: [10.3897/rio.10.e124168](https://doi.org/10.3897/rio.10.e124168).

Books and Chapters

- 1 Rubén J. García-Hernández et al. "Event-Driven, Time-Constrained Workflows." In *HPC, Big Data, and AI Convergence Towards Exascale*, CRC Press, 2022. DOI: [10.1201/9781003176664-7](https://doi.org/10.1201/9781003176664-7).
- 2 Thierry Goubier et al. "Earthquake and Tsunami Workflow Leveraging the Modern HPC/Cloud Environment in the LEXIS Project." In *Advances in Networked-based Information Systems*, Springer, 2020. DOI: [10.1007/978-3-030-29029-0_21](https://doi.org/10.1007/978-3-030-29029-0_21).
- 3 Thierry Goubier et al. "Real-Time Model of Computation over HPC/Cloud Orchestration - The LEXIS Approach." In *Complex, Intelligent and Software Intensive Systems*, Springer, 2021. DOI: [10.1007/978-3-030-50454-0_24](https://doi.org/10.1007/978-3-030-50454-0_24).
- 4 Marek Lampart and Tomáš Martinovič. "CML-Tent Model Chaotic Behavior with Respect to the State and Coupling Parameters." In *Chaos and Complex Systems*, Springer, 2020. DOI: [10.1007/978-3-030-35441-1_2](https://doi.org/10.1007/978-3-030-35441-1_2).
- 5 Tomáš Martinovič et al. "Automated Application of Inventory Optimization." In *Computer Information Systems and Industrial Management*, Springer, 2016. DOI: [10.1007/978-3-319-45378-1_21](https://doi.org/10.1007/978-3-319-45378-1_21).
- 6 Tomáš Martinovič et al. "Inventory Optimization Model Parameter Search Speed-Up Through Similarity Reduction." In *Soft Computing Applications*, Springer, 2021. DOI: [10.1007/978-3-030-51992-6_9](https://doi.org/10.1007/978-3-030-51992-6_9).
- 7 Tomáš Martinovič et al. "Sequence Analysis for Relationship Pattern Extraction." In *Computer Information Systems and Industrial Management*, Springer, 2019. DOI: [10.1007/978-3-030-28957-7_29](https://doi.org/10.1007/978-3-030-28957-7_29).
- 8 Lukáš Rapant et al. "Traffic Speed Prediction Using Hidden Markov Models for Czech Republic Highways." In *Agent and Multi-Agent Systems: Technology and Applications*, Springer, 2016. DOI: [10.1007/978-3-319-39883-9_15](https://doi.org/10.1007/978-3-319-39883-9_15).

Scientific Projects

Primary investigator

- 2026-2030 ■ **PeatGenie**, project details to be confirmed.
Work Package Leader.
- 2023-2027 ■ **DTO-BioFlow: Integration of biodiversity monitoring data into the Digital Twin Ocean.**
DOI: [10.3030/101112823](https://doi.org/10.3030/101112823). *Task Leader.*
- 2022-2025 ■ **BioDT: Biodiversity Digital Twin for Advanced Modelling, Simulation and Prediction Capabilities.**
DOI: [10.3030/101057437](https://doi.org/10.3030/101057437). *Work Package Leader.*
- 2023-2026 ■ **Apollo data s.r.o. - Application 1 - EyeERP - a comprehensive ERP system for eye clinics with the MojeOči mobile application for performing and analyzing eye tests.**
E122_002/0000436.

International

- 2026-2029 ■ **Czech AI Factory.**
DOI: [10.3030/101314740](https://doi.org/10.3030/101314740). *Work Package Co-Leader.*
- 2025-2028 ■ **LAIFS: LUMI AI Factory - Service Center.**
DOI: [10.3030/101234208](https://doi.org/10.3030/101234208).
- 2025-2027 ■ **InnovAlte Slovakia: Illuminating Pathways for AI-Driven Breakthroughs.**
09I02-03-V01-00029.
- 2023-2026 ■ **EXA4MIND: EXtreme Analytics for MINing Data spaces.**
DOI: [10.3030/101092944](https://doi.org/10.3030/101092944). *Task Leader.*
- 2021-2024 ■ **ACROSS: HPC Big Data Artificial Intelligence Cross Stack Platform Towards Exascale.**
DOI: [10.3030/955648](https://doi.org/10.3030/955648). *Task Leader.*
- 2020-2022 ■ **EuroCC: National Competence Centres in the framework of EuroHPC.**
DOI: [10.3030/951732](https://doi.org/10.3030/951732).
- 2019-2021 ■ **LEXIS: Large-scale EXecution for Industry & Society.**
DOI: [10.3030/825532](https://doi.org/10.3030/825532). *Task Leader.*

National

- 2026-present ■ **Národní Centrum Umělé Inteligence**, details to be confirmed.
- 2021-2023 ■ **SmartFleet - AI-based software for full use of electric vehicles in companies and maximizing their share in the vehicle fleet.**
EG20_321/0024896.
- 2019-2021 ■ **Contact-less detector for partial discharges activity in medium overhead voltage powerlines.**
TJ02000031.
- 2016-2020 ■ **IT4Innovations excellence in science.**
LQ1602.

Contractual research

2016-2017 ■ IXPERTA s.r.o.

2015-2016 ■ K2 Atmitec s.r.o.

Teaching Experience

University Courses

since 2024 ■ **Libraries for parallel data processing.**

Tutor, teacher giving lectures.

Bachelor/Diploma Thesis

2026 ■ Petr Kladov, *Kolmogorov-Arnold Network analysis and comparison with classical neural networks*, VSB - Technical University of Ostrava.

Diploma Thesis - Supervisor.

2023 ■ Matěj Eliáš, *Tool for iterative visualization of selected protein characteristics in a web interface*, VSB - Technical University of Ostrava.

Bachelor Thesis - Supervisor.

2020 ■ Jan Pastorek, *Use of High-level Synthesis to accelerate application development to programmable logic*, VSB - Technical University of Ostrava.

Bachelor Thesis - Consultant.

Training Events

May 2026 ■ AI Confidential, Prague.

April 2026 ■ AI Confidential, Brno.

April 2026 ■ AI Confidential, Ostrava.

June 2025 ■ Python for Data Stewards, Ostrava.

April 2025 ■ Python for HPC, Online.

May 2024 ■ Parallel R, Online.

April 2023 ■ High Performance Data Analysis with R, Ostrava/Hybrid.

May 2022 ■ Data science with R and Python, Online.

September 2020 ■ Applied Image Segmentation using Intel Architecture, Online.

July 2020 ■ Deep Learning on Intel Architecture, Online.

May 2020 ■ Deep Learning on Intel Architecture, Online.

July 2019 ■ Intel AIDC Summit, Munich.

June 2019 ■ Hands-on Developer Workshop for Artificial Intelligence, Milan.

April 2019 ■ Data science with R and Python - Ostrava.

Summer School Lectures

March 2025 ■ **Organising committee and lectures**, BioDT School, Lecce.

August 2024 ■ **Data Science with R**, EUMaster4HPC, Summer School, Ostrava.

June 2023 ■ **Data Science with R**, SCTrain Summer School, Ostrava.

Miscellaneous Experience

Awards

- June 2019 ■ **Best Poster Award**, *HPC Oriented Algorithm for Computation of Recurrence Quantitative Analysis*, ISC High Performance 2019.
- May 2019 ■ **1st Place in The Competition for the Best Dissertation Defended in 2018, Topic: Information Technology and Electrical Engineering.**
Tools for time series analysis of nonlinear dynamical systems, VSB - Technical University of Ostrava and Consortium PROGRES 3.

Scientific Reviews

- since 2026 ■ Frontiers in Plant Science.
- since 2026 ■ SKV - Systém kvalitních výsledků.
- since 2025 ■ Journal of Science and Technology.
- since 2024 ■ Ecological Informatics.
- since 2024 ■ Ecological Modelling.
- since 2019 ■ International Journal of Computer Mathematics.
- since 2018 ■ The Journal of Supercomputing.
- since 2018 ■ Mathematical Methods in the Applied Sciences.
- since 2016 ■ CISIM - International Conference on Computer Information Systems and Industrial Management.

Participation in Organisations and Conference

- since 2024 ■ ETP4HPC SRA7 HPC / AI Applications and Workflows.
- since 2020 ■ BDVA Task Force Finance.
- 2026 ■ TDWG 2026 co-convening session on *Digital Twins, Data Infrastructures*.
- since 2026 ■ ELIXIR CZ AI Positioning framework.

International Mobility

- June-November 2018 ■ Politecnico di Milano, Dipartimento di Elettronica, Informazione e Bioingegneria.

Other

- June 2024 ■ **Invited Panel Discussion.** Distributed Digital Twins: Engineering Challenges and Future Trends.